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Sketch of the microlocal

characterization of $\text{im}(K_\Sigma)$

Last time. Overview of microsupport, the FLTZ Skeleton,
Statement of the main theorem.

Today. Sketch the proof of the main theorem

Recall. For a fan (N, Σ) , the FLTZ Skeleton is the closed,
 $\mathbb{R}_{>0}$ -conic, subanalytic Lagrangian

$$\Lambda_{\Sigma} := \bigcup_{m \in M, \sigma \in \Sigma} (m + \sigma^{\perp}) \times (-\sigma)$$

$$\text{in } T^*M_{\mathbb{R}} \simeq M_{\mathbb{R}} \times N_{\mathbb{R}}.$$

> we always have $\text{im}(K_{\Sigma}) \subset \text{Sh}_{\Lambda_{\Sigma}}(M_{\mathbb{R}}) \subset \text{Cons}_{P_{\Sigma}}(M_{\mathbb{R}})$

Toric Mirror Symmetry. If (N, Σ) is (smooth) projective,
then

$$K_{\Sigma} : \text{QCoh}(X_{\Sigma}/T) \longrightarrow \text{Sh}_{\Lambda_{\Sigma}}(M_{\mathbb{R}})$$

is an equivalence.

Proof strategy

> The idea is rather simple.

Def Let $\mathcal{C}$ be a stable ∞ -category and $D \subset \mathcal{C}$ a stable subcategory.

(1) The right orthogonal to D is

$$D^\perp = \{c \in \mathcal{C} \mid \text{Map}_{\mathcal{C}}(d, c) = 0 \text{ for all } d \in D\}$$

(2) The left orthogonal to D is

$${}^\perp D = \{c \in \mathcal{C} \mid \text{Map}_{\mathcal{C}}(c, d) = 0 \text{ for all } d \in D\}$$

Note Both $D^\perp$ and ${}^\perp D$ are stable subcategories. Moreover, we have inclusions

$$D \subset {}^\perp(D^\perp) \quad \text{and} \quad D \subset ({}^\perp D)^\perp$$

Lem Let $\mathcal{C}$ be a stable ∞ -category and $i: D \hookrightarrow \mathcal{C}$ a fully faithful left adjoint where D is stable. Then

$$D = {}^\perp(D^\perp)$$

Upshot. In this situation, if $D^\perp = 0$, then since ${}^\perp(0) = \mathcal{C}$, the functor i is an equivalence.

Proof Strategy

(1) Construct for each $x \in M_{\mathbb{R}}$ a sheaf $w(D_x)$ such that for $F \in \mathrm{Sh}_{\wedge_{\Sigma}}(M_{\mathbb{R}})$, we have

$$\underline{\mathrm{Map}}(w(D_x), F)[\dim(M_{\mathbb{R}})] \simeq F_x$$

(2) Show that each $w(D_x)$ is in $\mathrm{im}(k_{\Sigma})$.

(3) Then by construction, every object of $\mathrm{im}(k_{\Sigma})^\perp$ has vanishing stalks, so $\mathrm{im}(k_{\Sigma}) = \mathrm{Sh}_{\wedge_{\Sigma}}(M_{\mathbb{R}})$.

> For (1) we need a more combinatorial collection of generators for $\mathrm{im}(k_{\Sigma})$.

A first observation

Ntn Given a presentable stable ∞ -category $\mathcal{C}$ and collection S of objects of $\mathcal{C}$, write $\langle S \rangle \subset \mathcal{C}$ for the closure of S under colimits and shifts.

> Combining what we've shown so far, we see

Prop. Let (N, Σ) be a smooth projective fan. Then

(1) $\text{im}(k_\Sigma) \subset \text{Sh}(M_{\mathbb{R}})$ is closed under colimits and shifts.

(2) We have $\text{im}(k_\Sigma) = \langle \omega_{m+\sigma v} \mid m \in M, \sigma \in \Sigma \rangle$

(3) $F \in \text{im}(k_\Sigma)$ if and only if for each $\sigma \in \Sigma$, we have

$$F * \omega_\sigma v \in \langle \omega_{m+\sigma v} \mid m \in M \rangle$$

(4) $\text{im}(k_\Sigma)$ contains the unit for convolution and is closed under convolution products.

Dualizing Sheaves of moment polytopes

Recall. Let (N, Σ) be a projective fan. A moment polytope for Σ is a polytope $P \subset M_{\mathbb{R}}$ such that Σ is the normal fan of P .

Combinatorial exercise. If P is a moment polytope for (N, Σ) , then

(1) For each $\sigma \in \Sigma$, there exists an $m \in M$ such that

$$P + \sigma^{\vee} = m + \sigma^{\vee}$$

(2) For each $\sigma \in \Sigma$ and $n \in M$, there are elements $(m_i)_{i \geq 0}$ of M such that

$$n + \sigma^{\vee} = \bigcup_{i \geq 0} m_i + \underbrace{i P}_{\text{scaling of } P \text{ by } i}.$$

scaling of
 P by i

Prop Let (N, Σ) be a (smooth) projective fan and let P be a moment polytope for Σ . Let $m \in M$ and $i \geq 1$. Then

(1) We have $\omega_{m+iP} \in \text{im}(k_\Sigma)$

(2) ω_{m+iP} is a compact object of $\text{Cons}_{P_\Sigma}(M_{\mathbb{R}})$, hence also compact in $\text{im}(k_\Sigma)$.

(3) we have $\text{im}(k_\Sigma) = \langle \omega_{m+iP} \mid m \in M, i \geq 1 \rangle$.

Proof

(1) we need to check that for each $\sigma \in \Sigma$, we have

$$\omega_{\sigma^\vee} * \omega_{m+iP} \in \langle \omega_{n+\sigma^\vee} \mid n \in M \rangle$$

For this, observe that $m+iP$ is a moment polytope for Σ . So there is an $n_i \in M$ such that $\sigma^\vee + (m+iP) = \sigma^\vee + n_i$

$$\omega_{\sigma^\vee} * \omega_{m+iP} \cong \omega_{\sigma^\vee + m + iP} \cong \omega_{\sigma^\vee + n_i}$$

(2) Recall that by exodromy, $\text{Cons}_{P_\Sigma}(M_{\mathbb{R}}) \cong \text{Fun}(P_\Sigma, \text{Sp})$. Moreover, a functor $P_\Sigma \rightarrow \text{Sp}$ is compact if and only if it takes finitely many values, all of which are compact spectra. So the claim follows from the fact that the polytope $m + iP$ is bounded and $\omega_{m+iP} \cong \mathbb{1}_{m+iP}[\dim(M_{\mathbb{R}})]$.

(3) By the previous exercise, we see that for each $n \in M$ and $\sigma \in \Sigma$,

$$\text{Colim}_{m+iP \subset n+\sigma v} \omega_{m+iP} \cong \omega_{n+\sigma v}$$

Since $\text{im}(k_\Sigma) = \langle \omega_{n+\sigma v} \mid n \in M, \sigma \in \Sigma \rangle$, the claim is immediate. $\square$

Twisted dualizing sheaves on divisors

Recall. If (N, Σ) is a fan and K is a separably closed field, then the group $\text{Div}_T(X_{\Sigma, K})$ of T -equivariant divisors can be described as

$$\text{Div}_T(X_{\Sigma, K}) \cong \lim_{\sigma \in \Sigma^{\text{top}}} M / \underbrace{(\sigma^\perp + nM)}_{M(\sigma)}$$

$$\cong \text{Ker} \left(\bigoplus_{\sigma \in \Sigma_{\text{top}}} M/M(\sigma) \rightarrow \bigoplus_{\sigma, \tau \in \Sigma_{\text{top}}} M/M(\sigma \cap \tau) \right)$$

Over S , we'll simply refer to the right-hand side as a divisor on X_Σ and denote a divisor by

$$D = (m_\sigma)_{\sigma \in \Sigma}$$

Observe. Since $\sigma^\perp \subset \sigma^\vee$, the subset $m_\sigma + \sigma^\vee \subset M_{\mathbb{R}}$ is well-defined.

Moreover,

$$m_\sigma + (\sigma \cap \tau)^\vee = m_\tau + (\sigma \cap \tau)^\vee$$

If $\tau \subset \sigma$, then

$$(m_\sigma + \sigma^\vee) + \tau^\vee = m_\tau + \tau^\vee$$

Construction. Let (N, Σ) be a projective fan and let $D = (m_\sigma)_{\sigma \in \Sigma}$ be a divisor. We claim that the objects

$$\left(\omega_{m_\sigma + \sigma^\vee} \in \text{Mod}_{\omega_{\sigma^\vee}} \text{Sh}(M_{\mathbb{R}}) \right)_{\sigma \in \Sigma}$$

determine an object

$$\omega(D) \in \lim_{\sigma \in \Sigma^{\text{op}}} \text{Mod}_{\omega_{\sigma^\vee}} \text{Sh}(M_{\mathbb{R}}) \cong \text{Sh}(M_{\mathbb{R}})$$

descent for

idempotent algs.

That is, there are compatible equivalences

$$\omega_{m_\sigma + \sigma^\vee} * \omega_{\tau^\vee} \xrightarrow{\sim} \omega_{m_\tau + \tau^\vee}$$

whenever $z < \sigma$. These follow from the identifications

$$(m_\sigma + \sigma^\vee) + z^\vee = m_z + z^\vee$$

> More formally, we can apply the functor $\Gamma_{M, R}$ to the subcategory

$$\{m_\sigma + \sigma^\vee \mid \sigma \in \Sigma\} \subset \text{Poly}(M)$$

We already saw that for $X, Y \in \text{Poly}(M, R)$ of top dimension, the natural map

$$\omega_X * \omega_Y \longrightarrow \omega_{X+Y}$$

from the lax monoidal structure is an equivalence.

Lem. For each divisor $D = (m_\sigma)_{\sigma \in \Sigma}$, we have $\omega(D) \in \text{im}(k_\Sigma)$

Proof.

We need to check that for each $\sigma \in \Sigma$, we have $\omega(D) * \omega_{\sigma^\vee}$ is in $\langle \omega_{m + \sigma^\vee} \mid m \in M \rangle$. But by definition, $\omega(D) * \omega_{\sigma^\vee} = \omega_{m_\sigma + \sigma^\vee}$. $\square$

Exercise. If $D = D_P$ is the divisor of a polytope $P \in M_{\mathbb{R}}$, show that

$$\omega(D) \cong \omega_P.$$

Remark. For divisors D and D' , there is a natural equivalence

$$\omega(D + D') \cong \omega(D) * \omega(D').$$

The divisor of a point

Recall. Let (N, Σ) be a fan. Recall that there is an isomorphism

$$\lim_{\sigma \in \Sigma^{\text{op}}} M/M(\sigma) \simeq C^{\text{PL}}(|\Sigma|, \mathbb{R})$$

$$= \left\{ \begin{array}{l} \text{PL functions } |\Sigma| \rightarrow \mathbb{R} \\ \text{that are integral on} \\ \text{lattice points} \end{array} \right\}$$

Moreover, such a PL-function is determined by the value on each ray $\rho \in \Sigma(1)$.

> Recall that we write v_ρ for the minimal integral vector on the ray ρ .

Def. Let (N, Σ) be a projective fan and $x \in M_{\mathbb{R}}$. Write D_x for the divisor corresponding to the PL function whose value on a ray $\rho \in \Sigma(1)$ is

$$\lfloor -\langle x, v_\rho \rangle \rfloor + 1 = \begin{array}{l} \text{smallest integer strictly} \\ \text{larger than } -\langle x, v_\rho \rangle. \end{array}$$

The probing sheaf at x is $\omega(D_x)$.

> Recall that we already know that $\omega(D_x) \in \text{im}(k_\Sigma)$.

Thm. For a (smooth) projective fan (N, Σ) and $F \in \text{Sh}_{N_\Sigma}(M_{\mathbb{R}})$, we have a natural equivalence

$$\underline{\text{Map}}(\omega(D_x), F)[\dim(M_{\mathbb{R}})] \simeq F_x.$$

Cor. We have $\text{im}(k_\Sigma) = \text{Sh}_{N_\Sigma}(M_{\mathbb{R}})$.

Idea of proof. Fix a moment polytope P for Σ .

(1) By the characterization of divisors corresponding to polytopes, for $i \gg 1$, the divisor $D_x + D_P$ also corresponds to a moment polytope. Moreover, iP is a moment polytope, so we can replace P by iP and WLOG assume that

$$D_x + D_P = D_{P_x}.$$

Here, if $D_x + D_P = (n_p)_{p \in \Sigma(1)}$, then

$$P_x = \bigcap_{\beta \in \Sigma(1)} \{m \in M_{\mathbb{R}} \mid \langle m, v_{\beta} \rangle \geq -n_{\beta}\}$$

(2) $\omega_P = \omega(D_P)$ is convolution-invertible so

$$\omega_{P_x} \simeq \omega(D_x + D_P) \simeq \omega(D_x) * \omega_P$$

and

$$\omega(D_x) \simeq \omega_{P_x} * \omega_P^{*(-1)},$$

so $\omega(D_x)$ is 'close to' the dualizing sheaf of a polytope

(3) Then

$$\underline{\text{Map}}(\omega(D_x), F) \simeq \underline{\text{Map}}(\omega(D_x) * \omega_P, F * \omega_P)$$

$$\simeq \underline{\text{Map}}(\omega_{P_x}, F * \omega_P)$$

$$\omega_{P_x} \simeq \mathbb{1}_{P_x^{\circ}}[\dim(M_{\mathbb{R}})]$$

$$\simeq \mathbb{1}_{(F * \omega_P)(P_x^{\circ})}[-\dim(M_{\mathbb{R}})].$$

Also observe that for any sheaf G , Since ω_P is convolution-invertible, for $\varepsilon > 0$ we have

$$F(x + \varepsilon P^\circ) \simeq \underline{\text{Map}}(\mathbb{1}_{x + \varepsilon P^\circ}, F)$$

$$\xrightarrow{\sim} \underline{\text{Map}}(\mathbb{1}_{x + \varepsilon P^\circ} * \omega_P, F * \omega_P)$$

$$\simeq \underline{\text{Map}}(\underbrace{\mathbb{1}_{x + \varepsilon P^\circ} * \mathbb{1}_{P^\circ}[\dim(M, \mathbb{R})]}_{\mathbb{1}_{x + (1+\varepsilon)P^\circ} \text{ by earlier}}, F * \omega_P)$$

$$\simeq (F * \omega_P)(x + (1+\varepsilon)P^\circ).$$

Relating this to F_x uses the theory of microsupports and the non-characteristic deformation lemma.

One considers the one-parameter family of polytopes (for a small ε)

$$P_{x,s} = sP_x + (1-s)(x + (1+\varepsilon)P)$$

to show that for ε sufficiently small and $F \in \text{Sh}_{\Lambda_\Sigma}(M_R)$,

$$F(P_x^\circ) \simeq F(x + (1+\varepsilon)P^\circ)$$

Then

$$(F * \omega_P)(P_x^\circ) \simeq F(x + \varepsilon P^\circ)$$

So

$$\underline{\text{Map}}(\omega(D_x), F)(\dim(M_R)) \simeq F(x + \varepsilon P^\circ)$$

for any sufficiently small $\varepsilon > 0$.